Wave 6 — crescent [crescent_c4_seed] | Fitness -101.000 (SPG×A×(1+N)) | SPG 1.834 N/g | Est. weight 82.6 g (NOT LEGAL — reference only)

Detailed plate FEM (physics::fine DKT bending + T6 membrane) with per-element thickness field (taper, grinds, relief, lattice, fuller, flutes, texture). Capacities CERTIFIED = min over 0.25 / 0.125 mm² meshes; robust = min(20 °C at −0.1 mm stock, 100 °C yield)

AESTHETIC FLOOR: PASS
H 0.76 mm · EMD 0.000 · A 0.882

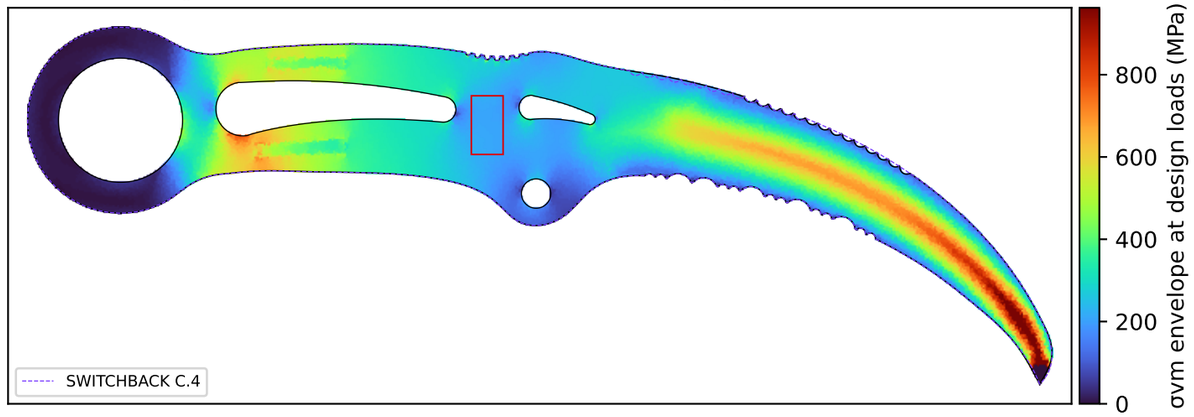
BLADE ≥ 3.375 in on all three: PASS
min 3.426 in

HARD GATES: FAIL
Holder envelope: best oriented box 200.5 x 100.2 > 200 x 100

FLOORS 180 / 260 / 1500 N (certified): FAIL
lat 152 N · ring 20983 N

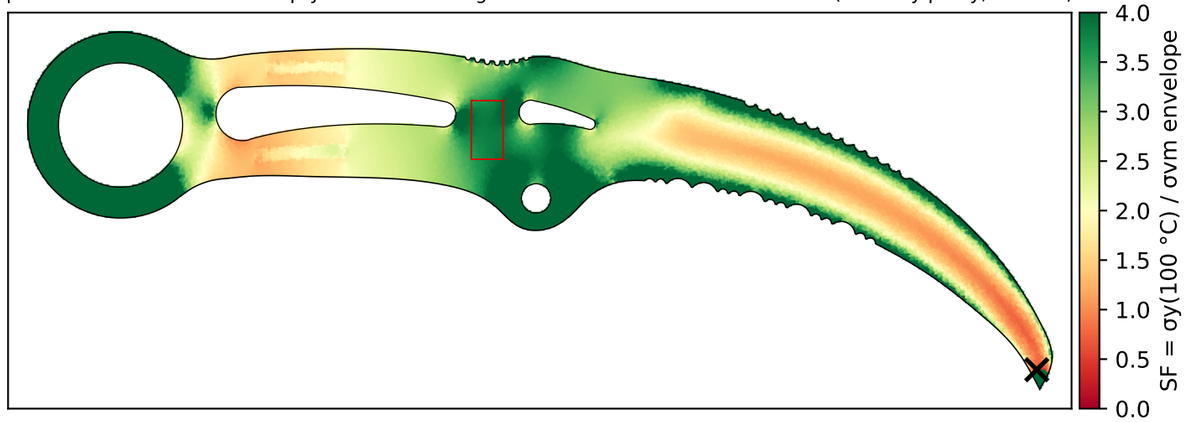
NOVELTY 0.000
(k-NN, aesthetic passers only)

von Mises envelope at design loads: lateral 260 N, pry 260 N, twist 1.26 N·m, ring pull 1500 N (4 mm tip-load zone excluded)

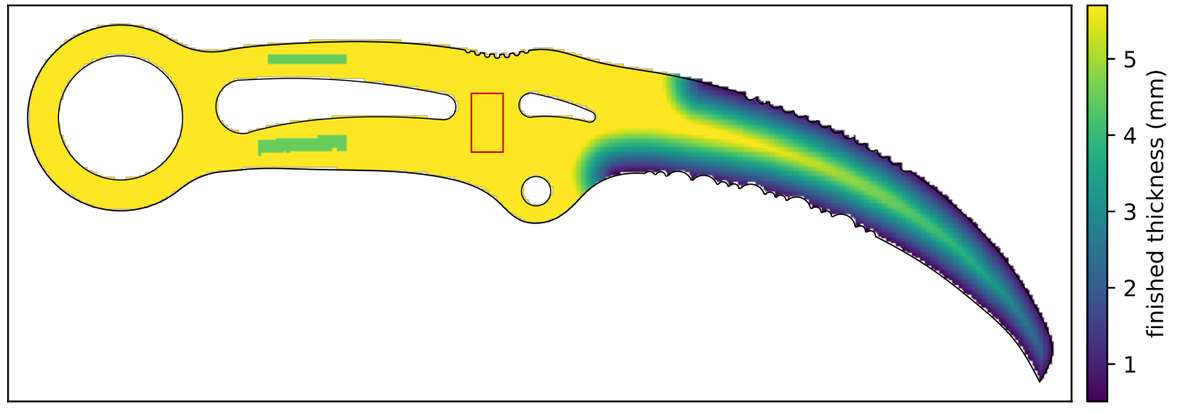


tip→heel	3.806 in (96.7)
horizontal reach	3.426 in (87.0)
edge arc inside / spine	3.983 / 3.963 in
grip / stock / Ti	60.5 mm / 5.7 mm / Ti-6Al-4V
weight (finished / blank)	82.6 g / 101.5 g
lateral 20 °C / robust	166 / 152 N
SF hiker 180 / hunter 260	0.84 / 0.58
ring pull 20 °C / robust (SF)	23008 / 20983 N (14.0)
tip pry 20 °C / twist 20 °C	719 N / 0.58 N·m
SF 100 °C lat / pry / ring / twist*	0.58 / 2.97 / 15.74 / 0.42
governing lateral at	(188, -50) mm
holder box (oriented)	200.5 × 100.2 mm @ -8.5°
aesthetic A / H / EMD	0.882 / 0.76 mm / 0.000
straight run / min blend R	16.5 / 13.3 mm
novelty / SPG	0.000 / 1.834 N/g

safety factor at 100 °C, envelope of all four cases (min 0.42 at x=187, y=-50 mm)
per case: lateral 260 N 0.58 · pry 260 N 2.97 · ring 1500 N 15.74 · twist 1.26 N·m 0.42 (advisory proxy, no floor)



thickness field from 5.7 mm stock (6.35 mm plate): removed relief 0 / lattice 0 / fuller 0 / flutes 97 / texture 0 mm³; min web 3.69 mm



novelty / QD map (all evaluations of the smoke run)

